



SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105.

DSA0307-Natural Language Processing

Stemming vs Morphological Parsing: A Comparative Study

Presented By:

S. Krishna Chaithanya (192424287)

Supervisor:

Dr.K.ANABAZHAGAN

INTRODUCTION

- Natural Language Processing (NLP) enables computers to understand and process human language.
- Text preprocessing is an essential step before applying NLP techniques.
- The main goal of preprocessing is to simplify text while preserving its meaning.
- Stemming and Morphological Parsing are two common word normalization techniques.
- Both methods reduce words to their base form but use different approaches.
Stemming is a rule-based approach that removes prefixes or suffixes.
- Morphological Parsing analyzes the grammatical structure of words.
- These techniques improve the efficiency and accuracy of NLP applications.

Stemming

What is Stemming?

- Stemming is the process of reducing words to their root or stem.
- It removes prefixes and suffixes using predefined rules.
- The generated stem may not always be a valid dictionary word.
- It does not consider grammar or word meaning.
- Mainly used for fast text processing.

EX:Playing → Play

Played → Play

Advantages

- Simple and easy to implement.
- Faster processing.
- Reduces storage space.
- Improves search engine performance.
- Suitable for large datasets.

Disadvantages

- May produce incorrect stems.
- Does not understand grammar.
- Different words may be reduced incorrectly.
- Lower linguistic accuracy.

Morphological Parsing

What is Morphological Parsing?

Morphological Parsing is the process of analyzing the internal structure of words. It identifies the root word, prefixes, suffixes, and grammatical information such as tense, number, gender, and case.

EX:

PLAYING =PLAY+ING

PLAYED=PLAY+ED

Advantages

- High accuracy.
- Maintains word meaning.
- Supports complex NLP tasks.
- Better language understanding.
- Suitable for multilingual applications.

Disadvantages

- Computationally expensive.
- Requires language-specific rules.
- More complex implementation.
- Slower than stemming.

Comparative Study

Stemming

- Reduces words to their stem by removing prefixes or suffixes.
- Uses simple rule-based algorithms.
- Fast and computationally efficient.
- Does not consider grammar or word meaning.
- May produce non-dictionary words.
- Suitable for search engines and information retrieval.

Morphological Parsing

- Analyzes the internal structure of words.
- Identifies prefixes, root words, and suffixes.
- Considers grammatical and linguistic information.
- Produces meaningful root words.
- More accurate but computationally complex.
- Suitable for machine translation, chatbots, and grammar checking.

Applications

Applications of Stemming

- Search Engines
- Google Search
- Information Retrieval
- Document Classification
- Text Mining
- Spam Filtering
- Keyword Matching

Applications of Morphological Parsing

- Machine Translation
- ChatGPT and Virtual Assistants
- Speech Recognition
- Grammar Checking
- Sentiment Analysis
- Text Summarization
- Named Entity Recognition

Conclusion

Conclusion

- Both Stemming and Morphological Parsing are essential preprocessing techniques in NLP.
- Stemming is simple, fast, and suitable for applications where speed is more important than linguistic accuracy.
- Morphological Parsing provides a deeper understanding of word structure and preserves grammatical information, making it ideal for advanced NLP tasks.
- The selection of either technique depends on the application's requirements, available computational resources, and desired level of language understanding.

Future Scope

- Hybrid NLP models combining stemming and morphological parsing.
- Improved multilingual language processing.
- AI-powered text understanding systems.
- Better support for low-resource languages.

*Thank
you!*